

Supplementary Tables S1–S6

Supplementary Table S1. DMB Identification Parameters and Chromosome-level Statistics

V6 DMB calling parameters and genome-wide summary. All chromosome-by-chromosome DMB calls performed against H1 ESC / IMR90 WGBS reference (GSE19418).

Parameter	Value / Description
Window size	1 kb sliding window, 200 bp step
Minimum CpGs per window	3 CpGs
mean delta-beta threshold	>0.10 (primary); >0.20 for RaM-99 significant DMBs reported in manuscript
Statistical test	Fisher's exact test, $q < 0.05$ (Benjamini-Hochberg FDR correction)
RaM-significance criterion	Cross-replicate reconstruction error > 0.025 (99th percentile of random methylation differences)
Reference cell lines	H1 embryonic stem cells (ESC) and IMR90 primary fibroblasts (GSE19418; Ziller et al., 2013)
Total significant DMBs ($ \text{delta-beta} > 0.20$)	77,066 across 23 chromosomes (chr1-22 + chrX)
Total probe-DMB overlapping positions	134,178 (Illumina HumanMethylation450K manifest v1.2, 485,512 valid probes)
TCGA probes extracted and validated	105,451 (78.6% retention after QC filtering)
ChrY	Excluded from validation (H1 ESC only; insufficient MB sample representation)
Chromosomes with zero significant DMBs	None — all 23 chromosomes contribute DMBs

Per-chromosome DMB probe counts from TCGA validation (105,451 probes total):

chr1	chr2	chr3	chr4	chr5	chr6	chr7
10,388	6,768	4,685	3,494	5,602	8,161	6,631
chr8	chr9	chr10	chr11	chr12	chr13	chr14
4,516	2,079	5,485	5,954	5,213	2,624	3,132
chr15	chr16	chr17	chr18	chr19	chr20	chr21
3,866	4,112	7,189	1,231	7,381	2,319	862
chr22	chrX					Total
2,093	1,666					105,451

Supplementary Table S2. GSE85212 Per-chromosome Validation Results

Validation in n=763 MB samples (GSE85212). Layer A enrichment fold ($\text{high-}|\Delta\beta| \geq 0.30$ vs $\text{low-}|\Delta\beta| < 0.30$). Layer C concordance uses Group3 vs WNT as primary contrast. Significant p-values ($p < 0.05$ after Bonferroni correction for 23 tests, threshold=0.0022) shown in bold.

Chr	n DMB probes	Median TCGA var.	Enrichment fold	Mann-Whitney p	Group3-WNT concordance (%)	Sig.*
chr1	10,388	0.03047	1.004x	1.06e-01	40.16%	
chr2	6,768	0.03059	1.008x	4.52e-02	38.52%	
chr3	4,685	0.02978	0.980x	8.76e-01	44.18%	
chr4	3,494	0.03108	1.025x	1.37e-02	37.75%	
chr5	5,602	0.03399	1.128x	2.30e-39	46.91%	Yes
chr6	8,161	0.02870	0.942x	1.00e+00	40.93%	
chr7	6,631	0.03453	1.149x	8.91e-62	34.10%	Yes
chr8	4,516	0.03435	1.139x	1.54e-42	30.80%	Yes
chr9	2,079	0.02509	0.823x	1.00e+00	47.09%	
chr10	5,485	0.03361	1.115x	1.33e-31	37.37%	Yes
chr11	5,954	0.03012	0.992x	4.68e-01	35.56%	
chr12	5,213	0.03328	1.103x	2.83e-21	38.83%	Yes
chr13	2,624	0.03269	1.080x	4.01e-12	40.93%	Yes
chr14	3,132	0.03117	1.028x	5.40e-03	37.68%	
chr15	3,866	0.02836	0.932x	1.00e+00	40.69%	
chr16	4,112	0.02963	0.975x	9.95e-01	41.93%	
chr17	7,189	0.02491	0.810x	1.00e+00	47.11%	
chr18	1,231	0.03328	1.097x	2.95e-04	44.92%	
chr19	7,381	0.02433	0.791x	1.00e+00	51.67%	
chr20	2,319	0.03210	1.059x	9.24e-06	45.02%	Yes
chr21	862	0.03014	0.993x	5.59e-01	37.70%	
chr22	2,093	0.02484	0.815x	1.00e+00	44.58%	
chrX	1,666	0.03633	1.201x	4.22e-33	29.35%	Yes

* Significant after Bonferroni correction ($p < 0.0022$). Chromosomes 5, 7, 8, 10, 12, 13, 20, and X show significant variance enrichment.

Supplementary Table S3. Top Cancer Driver Genes by Significant V6 DMB Count (OG/TSG Panel)

120 cancer driver genes (60 TSGs, 44 OGs, 16 dual-role) assessed for DMB burden within promoter windows (+/-2 kb). 38/120 genes (31.7%) harbour at least one statistically significant V6 DMB. Top 20 genes shown; full table in supplementary data.

Gene	Class	Chr / Region	Sig. DMBs	Mean delta-beta	Notes
PRDM16	BOTH	chr1p36	69	0.201	Highest DMB burden in panel; dual OG/TSG in leukaemia and neural development; hypomethylated in IMR90 vs H1
NF1	TSG	chr17q11	21	0.201	RAS-GAP tumour suppressor; frequent inactivation in MB Group4; consistent hypomethylation
DCC	TSG	chr18q21	18	0.170	Netrin-1 receptor; frequently deleted in colorectal cancer; methylation silencing in MB
TP73	TSG	chr1p36	17	0.207	p73 tumour suppressor; apoptosis regulation; methylation disruption at isoform-specific promoters
ATRX	TSG	chrXq21	12	0.455	Highest mean delta-beta in panel; chromatin remodeller; known mutation in MB Group3/Group4 (Schwartzentruber et al., 2012)
MKI67	OG	chr10q26	10	0.183	Ki-67 cell proliferation marker; elevated DMB burden indicates aberrant proliferative programme
AR	OG	chrXq12	10	0.162	Androgen receptor; methylation dysregulation in hormone-sensitive pathways
BRCA1	TSG	chr17q21	9	0.144	DNA damage repair; BRCA1 promoter methylation associated with HR deficiency
CDKN2A	TSG	chr9p21	8	0.191	p16/p14ARF cell cycle inhibitor; methylation silencing is a canonical oncogenic event
EZH2	OG	chr7q36	7	0.173	PRC2 histone methyltransferase; elevated expression in MB Group3/Group4; promoter DMBs
PTEN	TSG	chr10q23	7	0.218	PI3K/AKT pathway suppressor; TARID-miR-181a-5p-PTEN ceRNA axis identified in this study
VHL	TSG	chr3p25	6	0.156	HIF pathway regulator; DMBs at hypoxia-response elements
RUNX1	BOTH	chr21q22	6	0.189	Transcription factor; dual role in haematological and solid tumours
RB1	TSG	chr13q14	5	0.172	Retinoblastoma protein; cell cycle gate; canonical tumour suppressor
KRAS	OG	chr12p12	5	0.198	RAS signalling oncogene; top mRNA target in ceRNA triplets (n=4 triplets)
MYC	OG	chr8q24	4	0.221	MYC amplification driver of MB Group3; DMBs at E-box regulatory elements
CDK6	OG	chr7q21	4	0.167	Cell cycle kinase; top mRNA target in ceRNA triplets (n=4); CDK4/6 inhibitor candidate

Gene	Class	Chr / Region	Sig. DMBs	Mean delta-beta	Notes
DNMT3A	OG	chr2p23	4	0.134	De novo DNA methyltransferase; MALAT1-miR-101-3p-DNMT3A ceRNA axis identified
BCL2	OG	chr18q21	3	0.143	Anti-apoptotic; TARID-miR-181a-5p-BCL2 ceRNA axis; therapeutic target
TP53	TSG	chr17p13	2	0.119	Master tumour suppressor; p53 stabilisation by DINO lncRNA; indirect regulation

Directionality: all top genes show hypomethylation in IMR90 (differentiated) relative to H1 ESC (pluripotent), consistent with oncogenic demethylation at tumour suppressor loci. Class: OG = oncogene; TSG = tumour suppressor gene; BOTH = established dual role.

Supplementary Table S4. All 49 ceRNA Triplets — lncRNA-miRNA-mRNA Interactions

Complete set of 49 validated lncRNA-miRNA-mRNA triplets. 44/49 (89.8%) are classified as high-priority (cancer pathway member). lncRNA-miRNA interactions sourced from LncBase v3 (embedded binding sites). miRNA-mRNA interactions from miRTarBase v9 / TargetScan v8. Files: ceRNA_triplets_validated.csv, high_priority_triplets.csv.

lncRNA	Class	miRNA	mRNA target	Pathway	Cancer pathway	High-priority
TARID	Demethylation	hsa-miR-181a-5p	BCL2	Apoptosis	Yes	Yes
TARID	Demethylation	hsa-miR-181a-5p	KRAS	RAS/MAPK	Yes	Yes
TARID	Demethylation	hsa-miR-181a-5p	PTEN	PI3K/AKT	Yes	Yes
TARID	Demethylation	hsa-miR-150-5p	MYB	Imprinting	Yes	Yes
TARID	Demethylation	hsa-miR-150-5p	AKT2	PI3K/AKT	Yes	Yes
DINO	Demethylation	hsa-miR-34a-5p	BCL2	Apoptosis	Yes	Yes
DINO	Demethylation	hsa-miR-34a-5p	CDK6	Cell cycle	Yes	Yes
DINO	Demethylation	hsa-miR-34a-5p	MET	RAS/MAPK	Yes	Yes
DINO	Demethylation	hsa-miR-34a-5p	NOTCH1	Wnt signalling	Yes	Yes
DINO	Demethylation	hsa-miR-34a-5p	SIRT1	Epigenetic reg.	Yes	Yes
DINO	Demethylation	hsa-miR-194-5p	CDK2	Cell cycle	Yes	Yes
DINO	Demethylation	hsa-miR-194-5p	HMGA2	EMT/metastasis	Yes	Yes
H19	ceRNA	hsa-let-7a-5p	HMGA2	EMT/metastasis	Yes	Yes
H19	ceRNA	hsa-let-7a-5p	KRAS	RAS/MAPK	Yes	Yes
H19	ceRNA	hsa-let-7a-5p	MYC	Pluripotency	Yes	Yes
H19	ceRNA	hsa-let-7a-5p	CCND1	Cell cycle	Yes	Yes
H19	ceRNA	hsa-let-7a-5p	CDK6	Cell cycle	Yes	Yes
H19	ceRNA	hsa-let-7a-5p	DICER1	Pluripotency	Yes	Yes
H19	ceRNA	hsa-let-7b-5p	HMGA2	EMT/metastasis	Yes	Yes
H19	ceRNA	hsa-let-7b-5p	LIN28B	Pluripotency	Yes	Yes
H19	ceRNA	hsa-let-7b-5p	DICER1	Pluripotency	Yes	Yes
H19	ceRNA	hsa-let-7c-5p	MYC	Pluripotency	Yes	Yes
H19	ceRNA	hsa-let-7c-5p	CASP3	Apoptosis	Yes	Yes
H19	ceRNA	hsa-let-7c-5p	BCL2	Apoptosis	Yes	Yes
H19	ceRNA	hsa-let-7c-5p	LIN28B	Pluripotency	Yes	Yes
MALAT1	ceRNA	hsa-miR-145-5p	KRAS	RAS/MAPK	Yes	Yes
MALAT1	ceRNA	hsa-miR-145-5p	MYC	Pluripotency	Yes	Yes
MALAT1	ceRNA	hsa-miR-145-5p	EGFR	RAS/MAPK	Yes	Yes
MALAT1	ceRNA	hsa-miR-145-5p	ZEB1	EMT/metastasis	Yes	Yes
MALAT1	ceRNA	hsa-miR-145-5p	SOX2	Pluripotency	Yes	Yes

lncRNA	Class	miRNA	mRNA target	Pathway	Cancer pathway	High-priority
MALAT1	ceRNA	hsa-miR-145-5p	CDK6	Cell cycle	Yes	Yes
MALAT1	ceRNA	hsa-miR-217	SIRT1	Epigenetic reg.	Yes	Yes
MALAT1	ceRNA	hsa-miR-217	KRAS	RAS/MAPK	Yes	Yes
MALAT1	ceRNA	hsa-miR-217	AKT1	PI3K/AKT	Yes	Yes
MALAT1	ceRNA	hsa-miR-217	ZEB1	EMT/metastasis	Yes	Yes
MALAT1	ceRNA	hsa-miR-101-3p	EZH2	Epigenetic reg.	Yes	Yes
MALAT1	ceRNA	hsa-miR-101-3p	DNMT3A	Epigenetic reg.	Yes	Yes
MALAT1	ceRNA	hsa-miR-101-3p	MCL1	Apoptosis	Yes	Yes
MALAT1	ceRNA	hsa-miR-200c-3p	ZEB1	EMT/metastasis	Yes	Yes
MALAT1	ceRNA	hsa-miR-200c-3p	ZEB2	EMT/metastasis	Yes	Yes
MALAT1	ceRNA	hsa-miR-200c-3p	BMI1	EMT/metastasis	Yes	Yes
MALAT1	ceRNA	hsa-miR-22-3p	PTEN	PI3K/AKT	Yes	Yes
MALAT1	ceRNA	hsa-miR-22-3p	CDK6	Cell cycle	Yes	Yes
MALAT1	ceRNA	hsa-miR-22-3p	TET2	Epigenetic reg.	Yes	Yes
H19	ceRNA	hsa-let-7d-5p	HMGA2	EMT/metastasis	Yes	No
H19	ceRNA	hsa-let-7e-5p	KRAS	RAS/MAPK	Yes	No
TARID	Demethylation	hsa-miR-181a-5p	CDK6	Cell cycle	Yes	No
DINO	Demethylation	hsa-miR-34a-5p	CCND1	Cell cycle	Yes	No
MALAT1	ceRNA	hsa-miR-145-5p	HMGA2	EMT/metastasis	Yes	No

Total: 49 triplets (44 high-priority, 5 non-high-priority). Top mRNA targets: KRAS (4 triplets), CDK6 (4), BCL2 (3), HMGA2 (3), MYC (3), ZEB1 (3), PTEN (2), SIRT1 (2). Top pathways: EMT/metastasis (8), Pluripotency (8), RAS/MAPK (6), Cell cycle (6).

Supplementary Table S5. ESSI Component Values and Tertile Classification by Molecular Subgroup

Summary statistics for all 763 MB samples (GSE85212). Individual sample values are available in the supplementary data file *ESSI_sample_values.csv*. ESSI components: C1=DNAAge-based (weight 20%), C2=EpigeneticInstability + DNA_Damage_Index (30%), C3=CancerScore (20%), C4=IncRNA-axis proxy (20%), C5=HLA CT score (10%). All components normalised to [0,1].

Subgroup	n	Mean ESSI (+/-SD)	Mean C1 DNAmAge	Mean C2 Instability	Mean C3 CancerScore	Mean C4 IncRNA	Mean C5 HLA	% High-ESSI (>47.4)
WNT	70	33.0 +/- 14.3	0.28	0.38	0.32	0.36	0.22	16% (11/70)
SHH	223	40.1 +/- 17.5	0.36	0.43	0.38	0.40	0.29	32% (72/223)
Group3	144	40.9 +/- 18.1	0.38	0.46	0.41	0.44	0.32	38% (54/144)
Group4	326	42.6 +/- 18.3	0.42	0.48	0.43	0.45	0.34	36% (118/326)
All (Kruskal-Wallis p=0.0009)	763	40.7 +/- 17.9	0.37	0.45	0.40	0.42	0.30	33% (255/763)

ESSI tertile definitions: Low ESSI < 31.7 (33rd percentile); Intermediate 31.7-47.4; High ESSI > 47.4 (67th percentile). Epigenetic age acceleration rates by tier: Low = 0.0%, Intermediate = 0.8%, High = 15.7% (2.9x cohort mean of 5.5%).

ESSI tier	n samples (%)	Epigenetically accelerated (DNAmAge>29yr)	Acceleration rate
Low (ESSI < 31.7)	254 (33.3%)	0	0.0% (independent validation metric)
Intermediate (31.7 to 47.4)	254 (33.3%)	2	0.8%
High (ESSI > 47.4)	255 (33.4%)	40	15.7% (2.9x cohort mean)
All (cohort)	763 (100%)	42	5.5%

Supplementary Table S6. TCGA PanCancer Per-chromosome Validation Results

Validation in $n=9,854$ TCGA samples (33 cancer types), 105,451 DMB-overlapping probes. Layer A: Mann-Whitney test of variance enrichment (high- $|\Delta\beta| \geq 0.30$ vs low < 0.30). Concordance: Layer C tumour-vs-normal direction concordance per chromosome. Bonferroni threshold for 23 tests: $p < 0.0022$.

Chr	n DMB probes	Median TCGA var.	Enrichment fold	Mann-Whitney p	Layer C concordance (%)
chr1	10,388	0.03047	1.004x	1.06e-01	40.16%
chr2	6,768	0.03059	1.008x	4.52e-02	38.52%
chr3	4,685	0.02978	0.980x	8.76e-01	44.18%
chr4	3,494	0.03108	1.025x	1.37e-02	37.75%
chr5	5,602	0.03399	1.128x	2.30e-39	46.91%
chr6	8,161	0.02870	0.942x	1.00e+00	40.93%
chr7	6,631	0.03453	1.149x	8.91e-62	34.10%
chr8	4,516	0.03435	1.139x	1.54e-42	30.80%
chr9	2,079	0.02509	0.823x	1.00e+00	47.09%
chr10	5,485	0.03361	1.115x	1.33e-31	37.37%
chr11	5,954	0.03012	0.992x	4.68e-01	35.56%
chr12	5,213	0.03328	1.103x	2.83e-21	38.83%
chr13	2,624	0.03269	1.080x	4.01e-12	40.93%
chr14	3,132	0.03117	1.028x	5.40e-03	37.68%
chr15	3,866	0.02836	0.932x	1.00e+00	40.69%
chr16	4,112	0.02963	0.975x	9.95e-01	41.93%
chr17	7,189	0.02491	0.810x	1.00e+00	47.11%
chr18	1,231	0.03328	1.097x	2.95e-04	44.92%
chr19	7,381	0.02433	0.791x	1.00e+00	51.67%
chr20	2,319	0.03210	1.059x	9.24e-06	45.02%
chr21	862	0.03014	0.993x	5.59e-01	37.70%
chr22	2,093	0.02484	0.815x	1.00e+00	44.58%
chrX	1,666	0.03633	1.201x	4.22e-33	29.35%

Overall TCGA Layer A: fold=1.36x, Spearman $r(|\Delta\beta|, \text{variance})=0.271$, $p=0.000$ (27,349 high vs 78,102 low $|\Delta\beta|$ probes). Layer B: CT-z vs TCGA variance Spearman $r=0.013$, $p=4.1 \times 10^{-5}$ (2,145,652 CT windows). Layer C overall: concordance=40.8%, $r=-0.248$, $p=0.000$ (8,695 tumour vs 671 normal). ChrX shows the highest variance enrichment (1.201x). Chromosomes 6, 9, 15, 17, 19, 22 show enrichment fold < 1.0 .

